

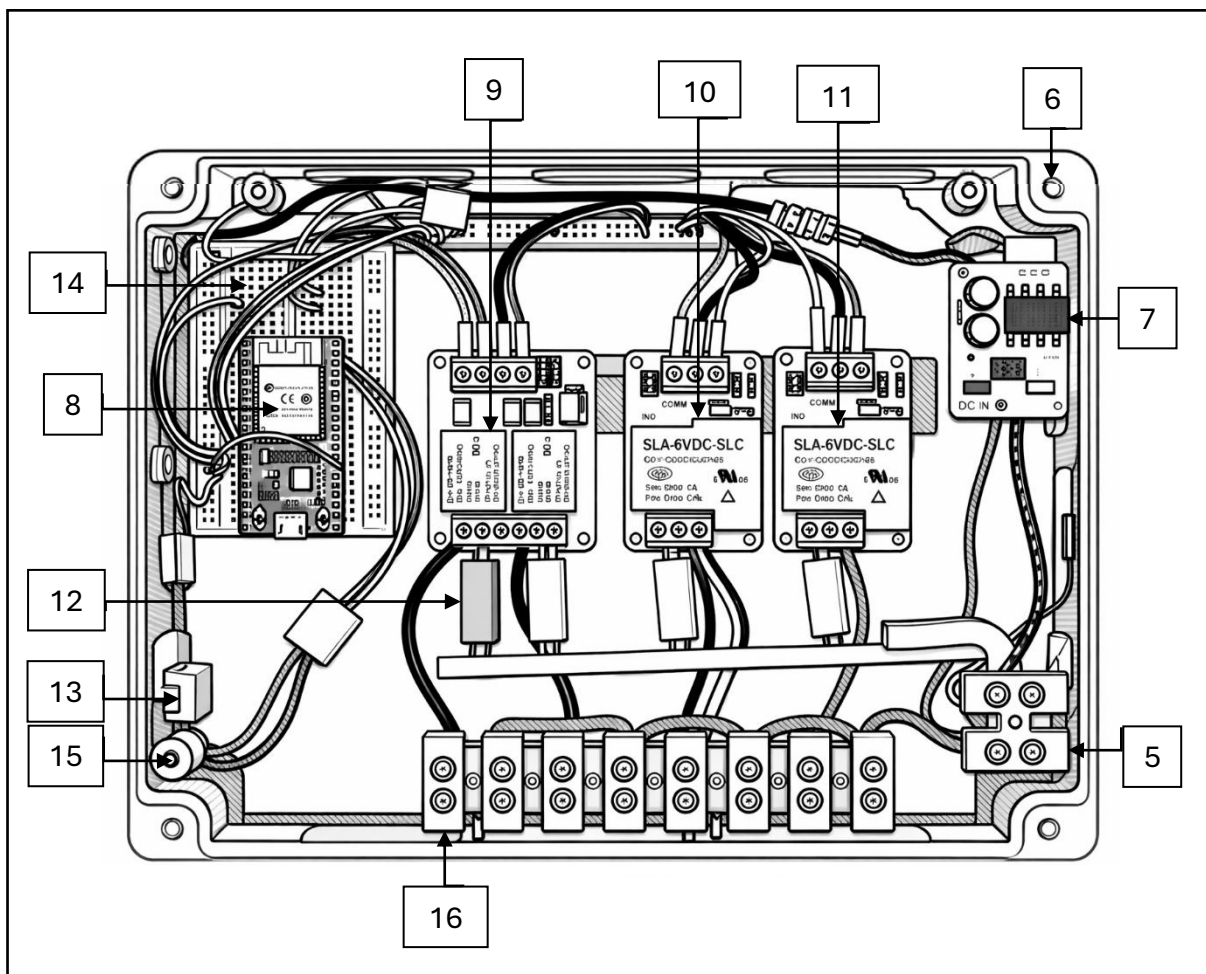
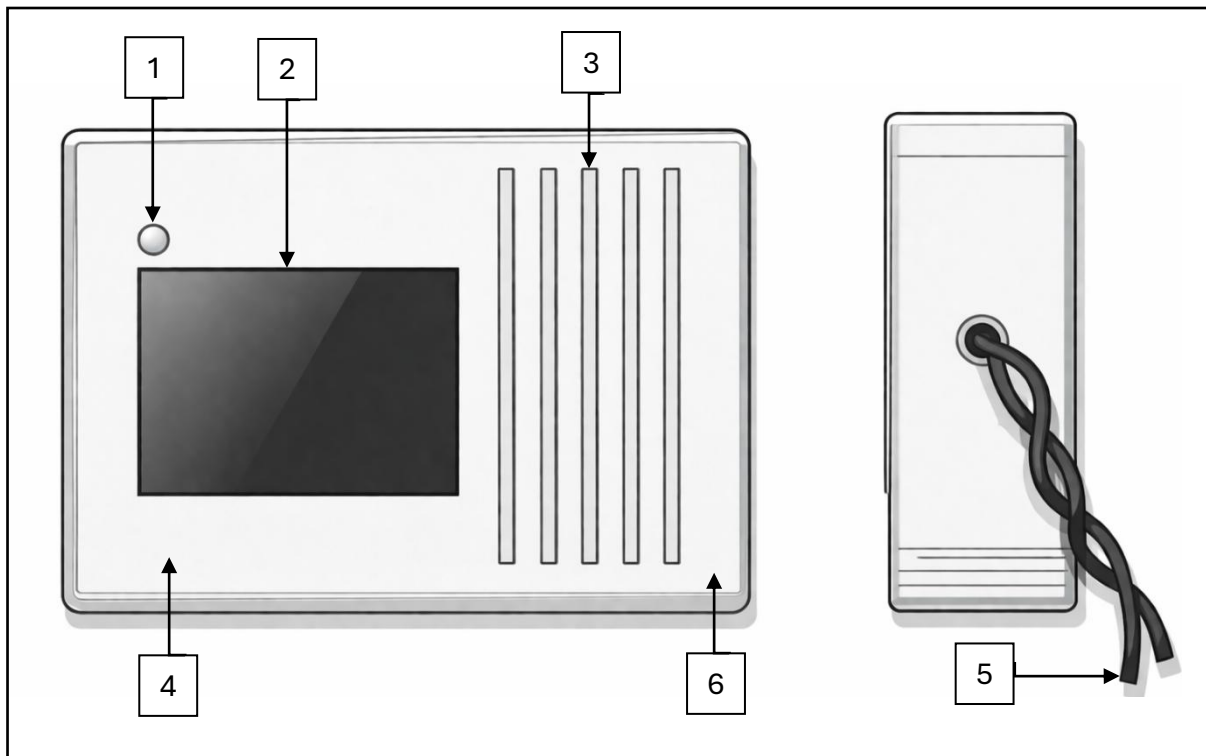


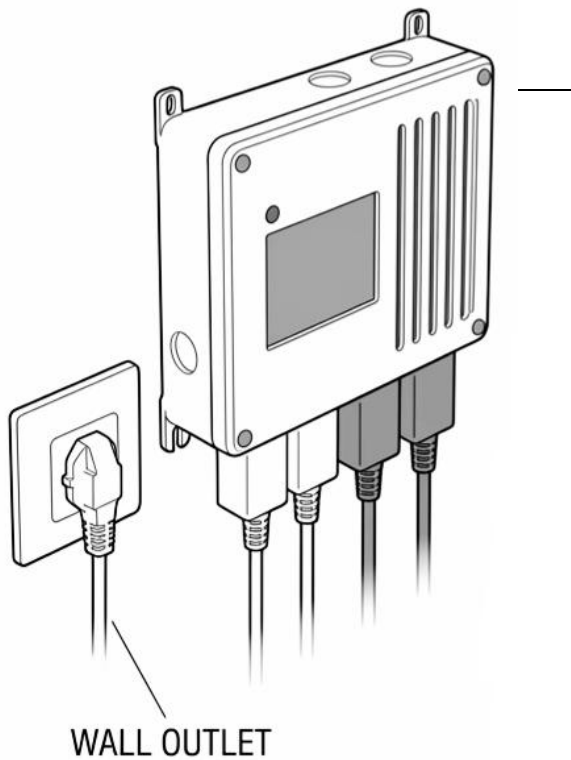
Home Automation Smart System



EN / Operating instructions

HASS Central Control & Sensor Network



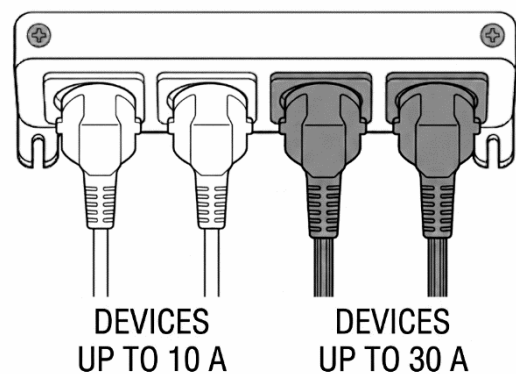
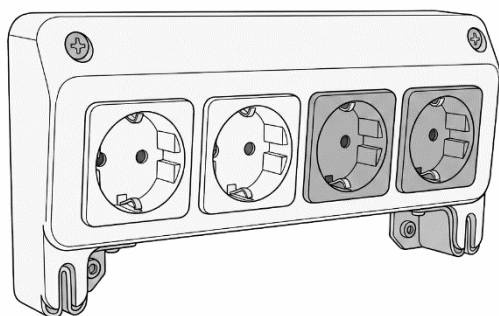
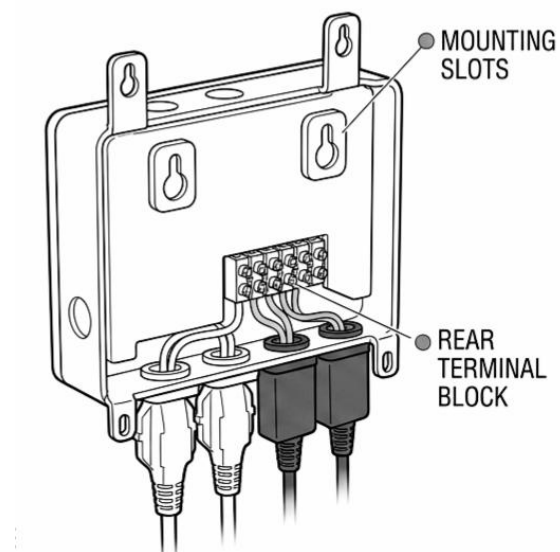


Control Panel

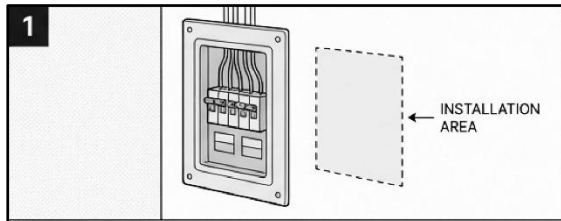
- [1] Wi-Fi Lamp
- [2] TFT Touchscreen Display
- [3] Ventilation openings
- [4] LED Indicators
- [5] Screw
- [6] Wiring Harness

System Interior

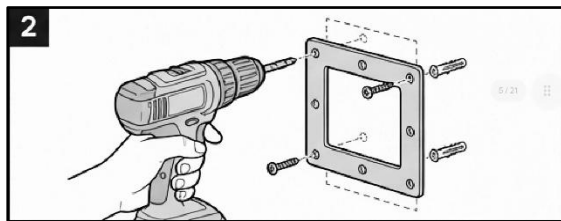
- [5] Adaptor
- [6] ESP32 Microcontroller Board
- [7] 2-Channel 10A Relay Module
- [8] 30A High-Power Relay
- [9] 30A High-Power Relay
- [10] DHT22 Temperature & Humidity Sensor
- [11] Fuses
- [12] Breadboard
- [13] Buzzer
- [14] Terminal devices Blocks
- [15] 2N2222 Transistors



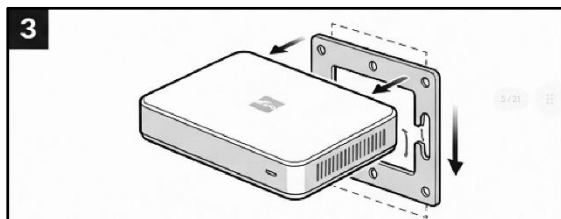
Step 1: Mounting the Hub



1. **Choose Location:** Mount the unit on a wall near your main electrical distribution board for easiest wire routing.

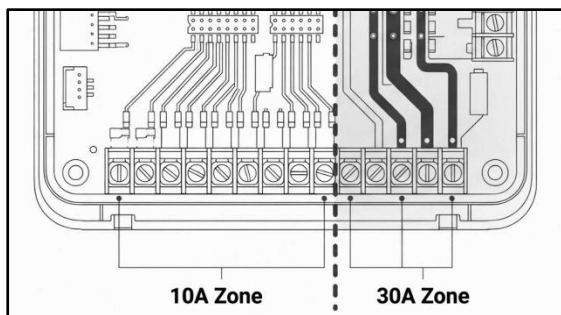


2. **Prepare the Wall:** Use the provided bracket as a template. Drill holes and insert the wall anchors.



3. **Secure the Unit:** Fasten the bracket with screws and slide the HASS hub securely into place.

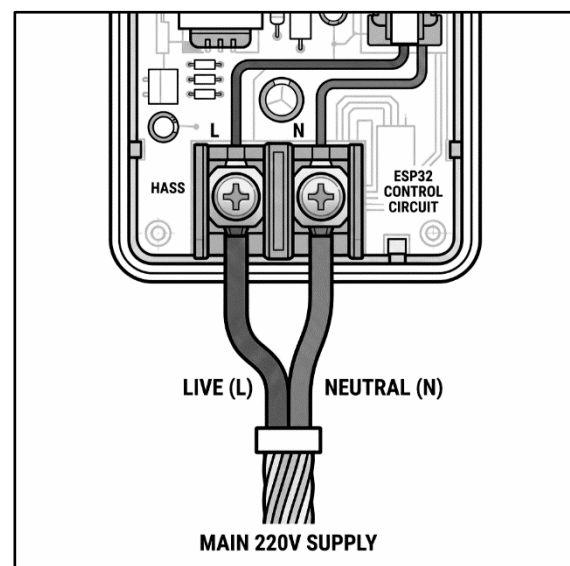
Step 2: Understanding the Terminal Blocks



The bottom of your HASS unit contains the relay outputs. This is where your home meets the smart system.

- **Left Terminals (10A):** STRICTLY for low-power electronics like lights and ceiling fans.
- **Right Terminals (30A):** Engineered for heavy loads like Air Conditioners (up to 2.25 HP) and water heaters.

Step 3: Powering the Hub

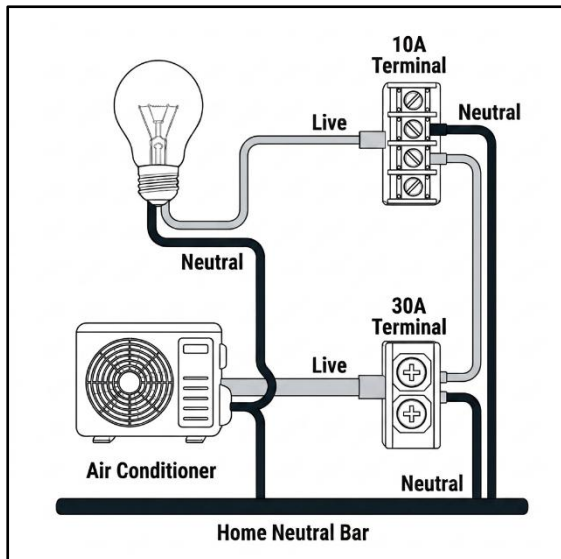


Safety Check: Confirm main breaker is OFF.

Action: Connect your home's main 220V Live wire and Neutral wire to the primary HASS input terminals to power the ESP32 control circuit.

Note: Use appropriately sized cable cross-sections (4mm² minimum for main lines) to ensure thermally stable operation.

Step 4: Connecting Household Appliances

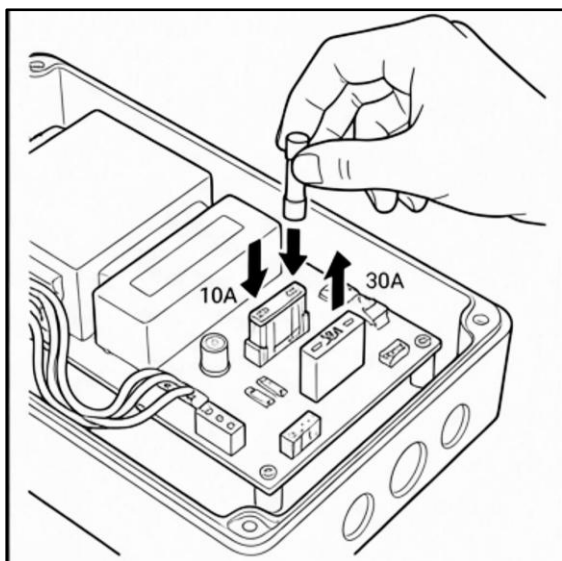


The "Live" Rule: Connect ONLY the "Live" wire of your appliance to the relay terminal.

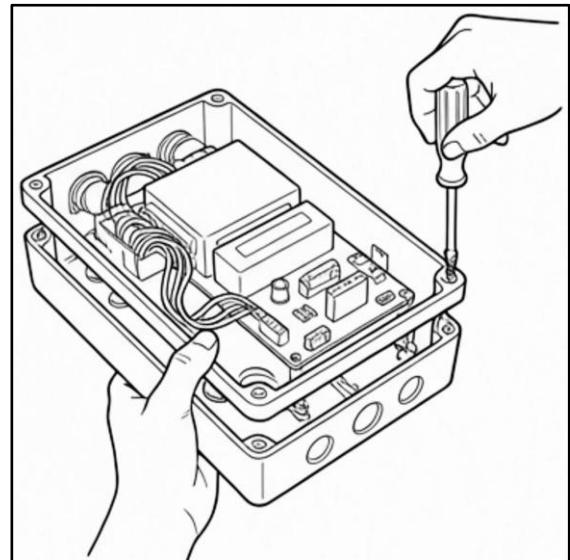
The "Neutral" Rule: Route all appliance "Neutral" wires directly to the neutral bar in your home electrical panel.

Galvanic Isolation: The internal relays safely separate the low-voltage brain from your 220V appliances.

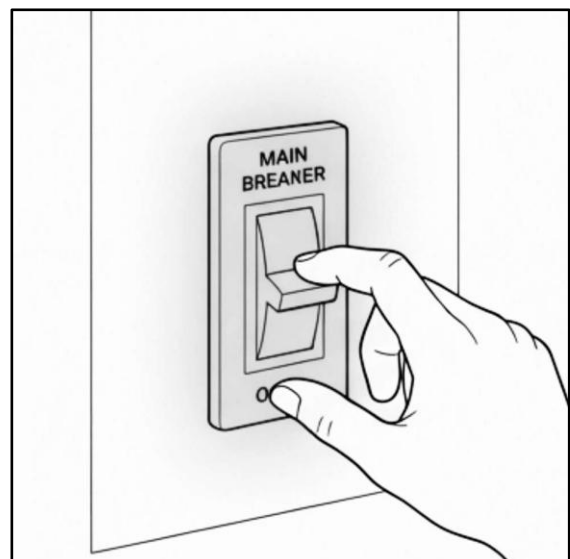
Step 5: Fuses & Final Safety Check



1. Insert Fuses: Place the 10A fuses into the low-load circuit holders, and the 30A fuses into the high-load circuit holders.



2. Close the Box: Secure the bottom cover of the HASS unit and tighten all screws.

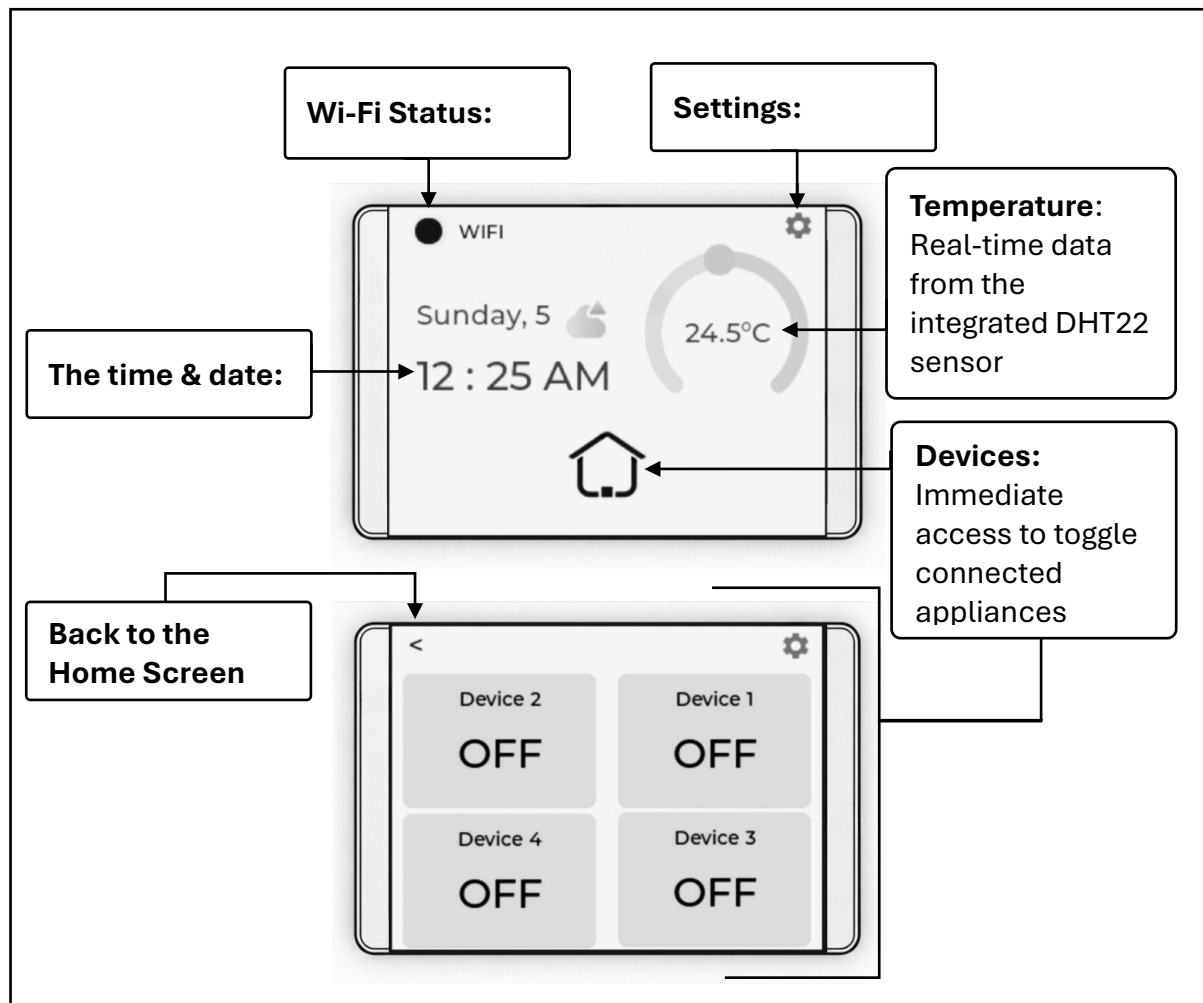


3. Restore Power: Turn the main power switch at your distribution board back ON.

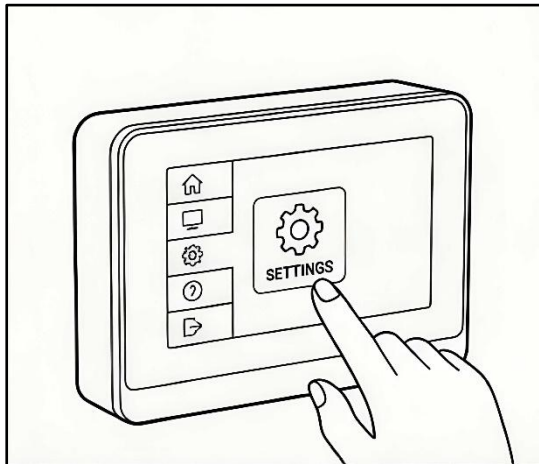
First Power-On & The Smart Display

Upon power restoration, the 7-inch touchscreen will boot up.

This is your local command center.

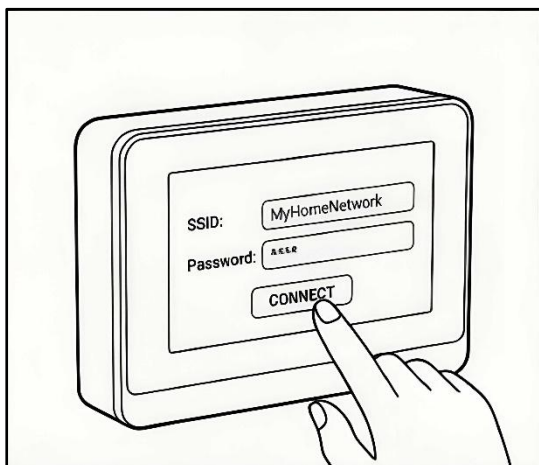


Connecting to Home Wi-Fi



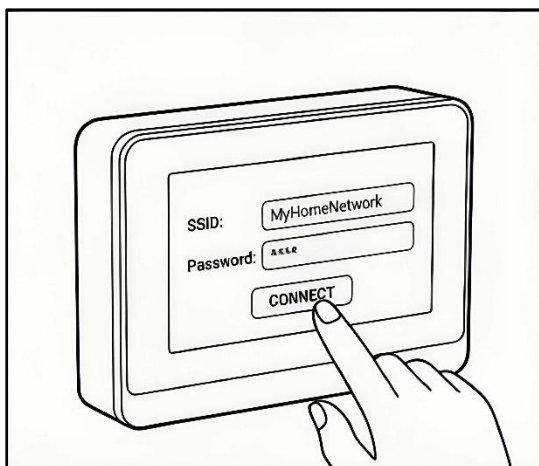
1.navigate

Tap the 'Settings' gear icon on the main TFT display.



2.Connection

Select your home network (SSID) and enter the password using the touch screen.



3.Confirm

Wait for the green indicator and the 'Connection Successful' prompt. Your HASS is now online.

Downloading the HASS App

Control your home from anywhere in the world, securely.



1.Open the App Store (iOS) or Google Play (Android).

2.Search for "HASS" or "Home Automation Smart System".

3.Download and install the application.

Account Creation & Device Discovery



1.Register

Open the app and create your personal secure account.



2.Discover

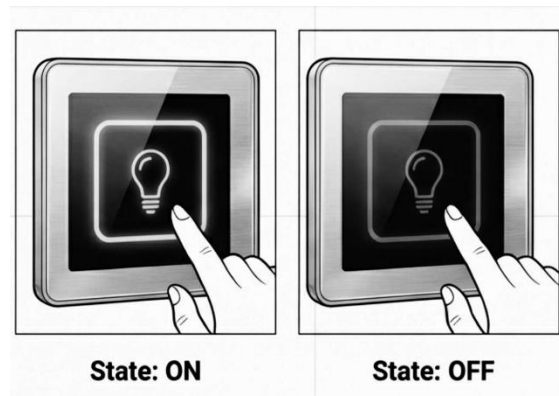
Ensure your phone is on the same Wi-Fi as the HASS hub and tap "Add New Device".



3.Pair

Enter the unique PAIR CODE shown on your wall unit's display to link the hardware to your account

Manual Control (Wall Screen)



The TFT display works completely independent of the internet. If your Wi-Fi goes down, your home still works.

Interaction: Simply tap any device icon to toggle power.

Feedback: The system provides instant visual feedback-Green for ON, Red for OFF-along with an audible buzzer confirmation.

Manual Control (Mobile App)

Your smartphone mirrors the wall unit in real-time. Changes made on the app reflect instantly on the wall display, and vice versa.

Live Status

View real-time device states and current room temperature.

Remote Toggle

Tap device icons to control power from anywhere via cloud integration.



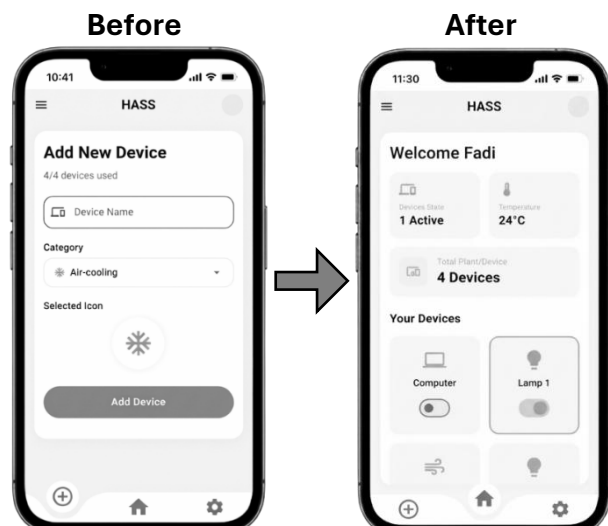
Renaming Your Sockets

Action: In the app settings, map your physical relay connections to custom names for intuitive

Steps:

1. Tap  Settings > Add/Edit Device.
2. Enter a custom Name.
3. Select an appropriate Category/Icon.
4. Tap Save.

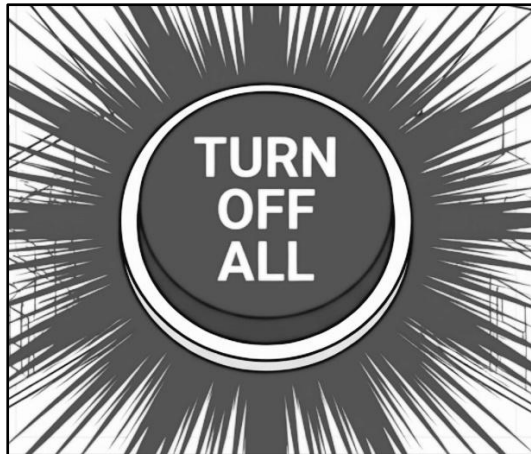
control



Function	Function Description	How to Activate
Manual Control ON/OFF Devices	Turn on or off any Device	Touch Screen (Without Wi-Fi) / Mobile Application (Wi-Fi)
Automation Mode IF-THEN Rules	Create automated conditions (If sensor is Triggered – Then turn on device)	Mobile Application (Wi-Fi)
Schedule Mode Weekly Routines	Create weekly routines: Lights , Thermostat , and more , automated by Time	Mobile Application (Wi-Fi)
Timer Mode Delayed Operation	Turn on or off Devices with a period of Time	Mobile Application (Wi-Fi)
Emergency Stop Turn Off All	Instant shutdown of all devices for safety	Mobile Application (Wi-Fi)
Monitoring Mode System Status	See real-time temperature, humidity & device status	Mobile Application (Wi-Fi)

The Emergency Stop Feature

Leaving in a hurry? Or suspect an electrical issue?



Function: The "Turn Off All" master switch provides an instant shutdown of every connected device for ultimate safety, peace of mind, and energy conservation. Available on both the TFT screen and mobile app.

Which Automation Should I Use ?

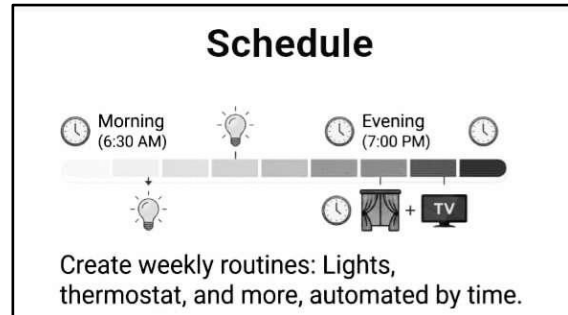


Timer Mod Mode (Delayed Operation)



Best for leaving a device on for a specific duration.

Example: Turn off the bathroom fan in 30 minutes.

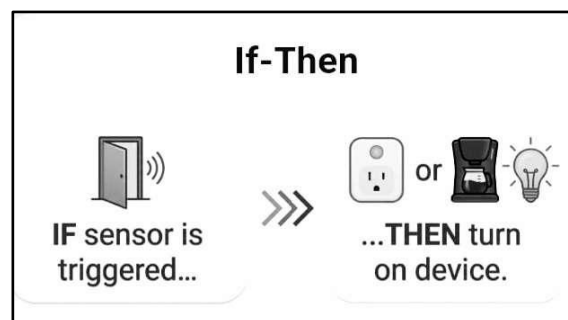


Schedule Mode (Weekly Routines)



Best for predictable daily habits.

Example: Turn on the coffee maker at 7:00 AM every weekday.



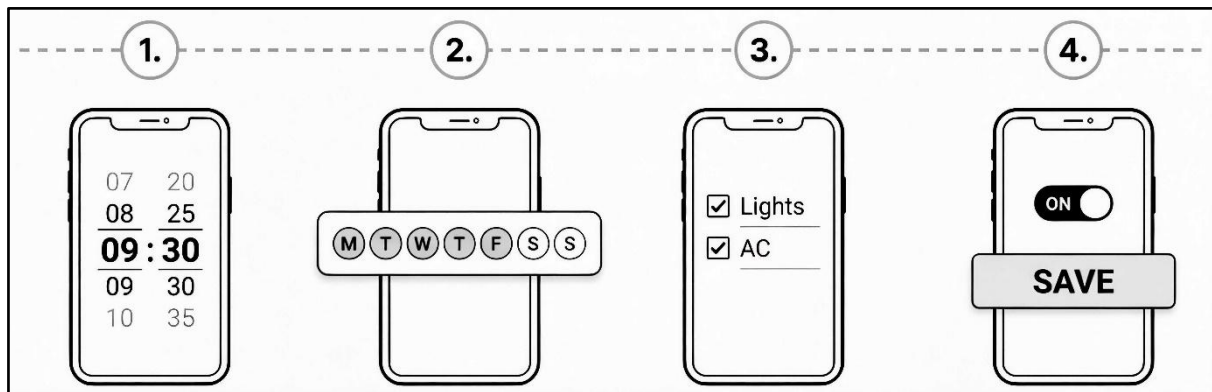
If-Then Mode (Sensor Triggered)



Best for environmental control.

Example: If the temperature exceeds 30°C, turn on the Air Conditioner.

Setting a Schedule



1. Pick Time:

Select the exact hour and minute.

2. Select Days:

Choose specific weekdays or weekends.

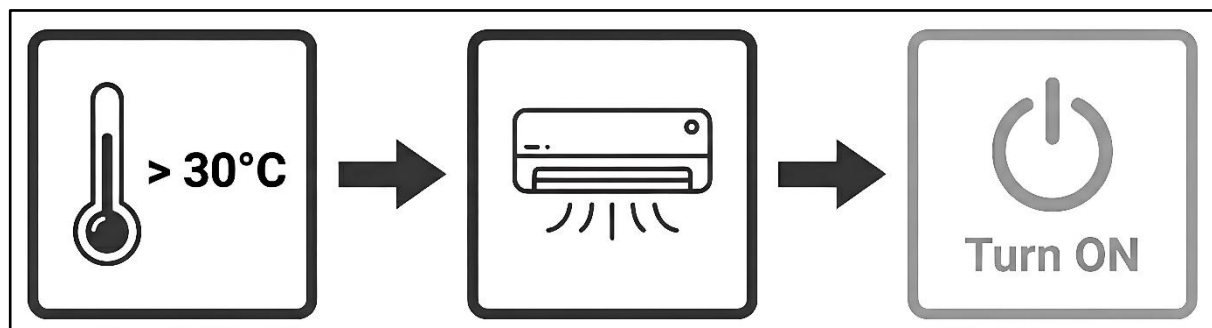
3. Choose Devices:

Select which appliances will be affected.

4. Set State & Save:

Choose whether they turn ON or OFF, then tap Save to activate the routine.

Setting an If-Then Rule



1. Set Condition:

Use the DHT22 sensor data to trigger an event (e.g., Temp $\geq 30^{\circ}\text{C}$).

2. Select Device:

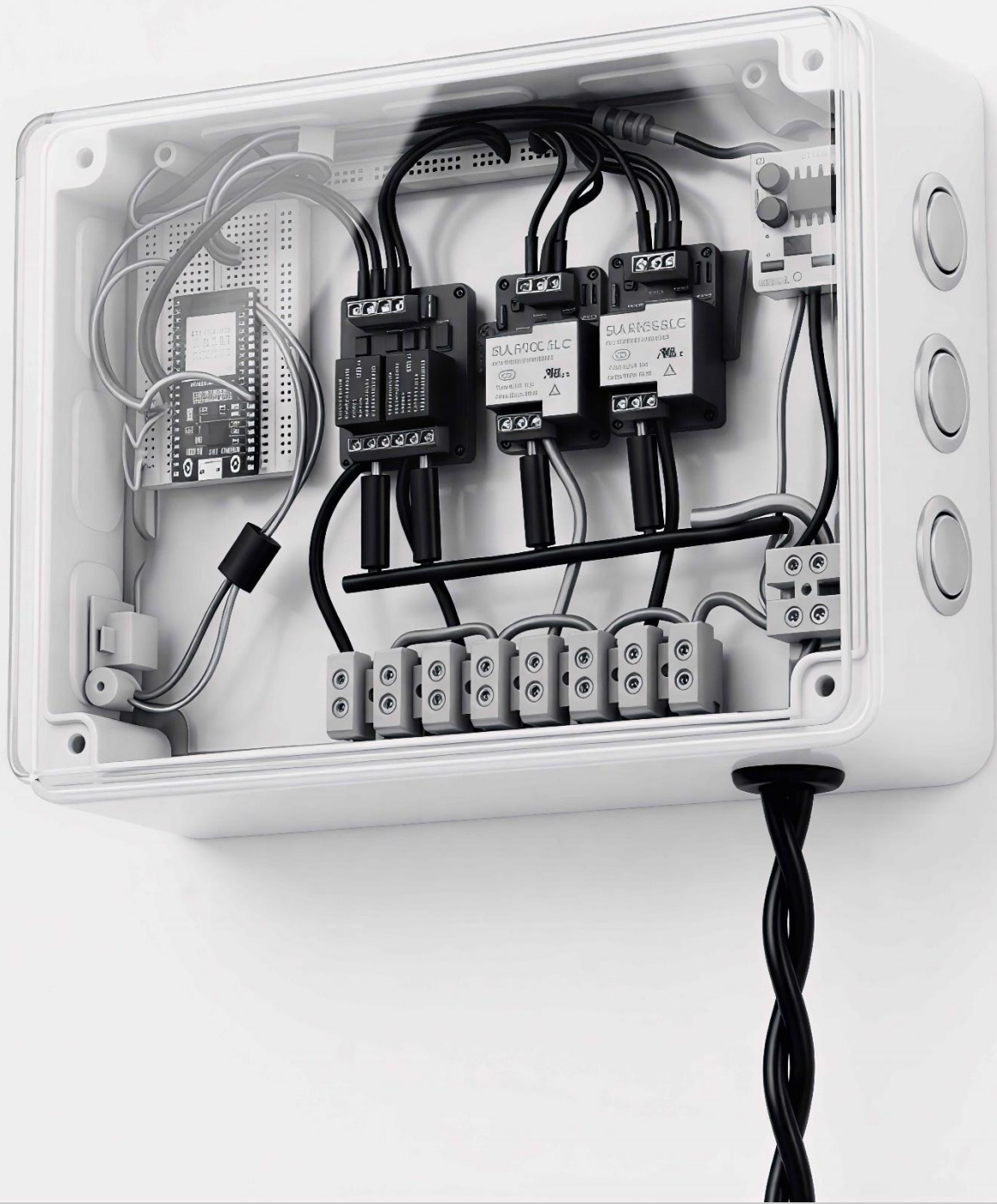
Choose the target appliance.

3. Choose Action:

Define what happens (Turn ON/OFF).

4. Save Rule:

The ESP32 brain will now monitor conditions and execute automatically, even without internet.



Thank you for your trust in us

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